Probability Basics

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The goal of probability theory is to study randomness and stochastic phenomena. Here, we overview some relevant concepts.

1 Probability Space and Sigma Algebra

A probability space is a mathematical construct that models a random experiment. It is defined by the triplet $(\Omega, \mathcal{F}, \mathbb{P})$ where:

- Ω: The sample space, which is the set of all possible outcomes.
- $\mathcal{F}$: The σ-algebra, which is a collection of subsets of Ω (called events) that includes Ω itself, is closed under complementation, and closed under countable unions. That is
  - $\Omega \in \mathcal{F}$
  - $A \in \mathcal{F} \implies A^C \in \mathcal{F}$
  - $A, B \in \mathcal{F} \implies A \cup B \in \mathcal{F}$ (true for any countable union).
- $\mathbb{P}$: The probability measure, which assigns a probability to each event in $\mathcal{F}$, satisfying the axioms of probability.

2 Probability Axioms

The probability measure $\mathbb{P}$ must satisfy the following axioms:

1. **Non-negativity**: For any event $A \in \mathcal{F}$, $\mathbb{P}(A) \geq 0$.
2. **Normalization**: $\mathbb{P}(\Omega) = 1$.
3. **Additivity**: For any countable sequence of disjoint events $A_1, A_2, \cdots \in \mathcal{F}$, $\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$.

Example: Coin Tossing - 0 for heads and 1 for tails

- $\Omega = \{0, 1\}$.
- $\mathcal{F} = \{\{0\}, \{1\}, \{0, 1\}, \varnothing\}$.
- Uniform measure $\mathbb{P}(A) = \frac{|A|}{|\Omega|}$.

$$\implies \mathbb{P}(\{0\}) = \frac{1}{2}, \mathbb{P}(\varnothing) = 0, \text{ etc.}$$

3 More about $(\Omega, \mathcal{F}, \mathbb{P})$

- Think of $\mathcal{F}$ as the superset of Ω, i.e., $\mathcal{F} = 2^{\Omega}$.
  But this is too big for general Ω, e.g., $\Omega = \mathbb{R}$.

- The smallest σ-algebra, B, containing the open subsets of $\mathbb{R}$ is called the Borel σ-algebra for $\mathbb{R}$. $A \in B$ is called a Borel set.
  E.g., $A = [0, 1]$ or $A = [0, 1)$.

4 Probability Space for Discrete and Continuous Random Variables

4.1 Discrete Random Variables

For a discrete random variable X, the probability space can be constructed by considering the sample space as the set of possible values that X can take, with $\mathcal{F}$ typically being the power set of Ω. The probability measure $\mathbb{P}$ is then defined by assigning probabilities to these values.

4.2 Continuous Random Variables

For continuous random variables, $\mathcal{F}$ is typically the Borel sigma algebra on $\mathbb{R}$, and $\mathbb{P}$ is defined using a probability density function (PDF) $f_X(x)$, such that for any event $A \in \mathcal{F}$:

$$\mathbb{P}(A) = \int_A f_X(x)\, dx.$$

5 Conditional Probability and Independence of Events

5.1 Conditional Probability

The conditional probability of an event A given another event B with $\mathbb{P}(B) > 0$ is defined as:

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Bayes' rule:

$$\mathbb{P}(A|B)\mathbb{P}(B) = \mathbb{P}(B|A)\mathbb{P}(A)$$

Total Probability:

$$\mathbb{P}(A) = \mathbb{P}(A|B)\mathbb{P}(B) + \mathbb{P}(A|B^C)\mathbb{P}(B^C) \geq \mathbb{P}(A|B)\mathbb{P}(B)$$

5.2 Independence of Events

Two events A and B are independent if $\mathbb{P}(A|B) = \mathbb{P}(A)$ or equivalently:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

6 Random Variables and Probability Distribution

6.1 Random Variables

A random variable X is a measurable function from a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ to the real numbers $\mathbb{R}$.

More generally, let $(\Omega, \mathcal{F})$ and $(\widetilde{\Omega}, \widetilde{\mathcal{F}})$ be two measurable spaces. Then $X : \Omega \to \widetilde{\Omega}$ is a random variable if $X^{-1}(\widetilde{A}) \in F$ for all $\widetilde{A} \in \widetilde{\mathcal{F}}$. In this case, X is called an $\mathcal{F}$-measurable map. Here, $X^{-1}(\widetilde{A})$ is the pre-image, i.e., $\{\omega \in \Omega \mid X(\omega) \in \widetilde{A}\}$.

6.2 Probability Distribution of X

Consider $(\Omega, \mathcal{F}, \mathbb{P})$ and $(\mathbb{R}, B(\mathbb{R}))$. Let $X : \Omega \to \mathbb{R}$ be $\mathcal{F}$-measurable. Then $\mathbb{P}_X(A) = \mathbb{P}(X^{-1}(A)) = \mathbb{P}(\{\omega \in \Omega \mid X(\omega) \in A\})$ is called the distribution of X.

Remark: Indeed, $\mathbb{P}_X$ is a probability measure.

- $\mathbb{P}_X(\mathbb{R}) = \mathbb{P}(X^{-1}(\mathbb{R})) = \mathbb{P}(\Omega) = 1$
- $\mathbb{P}_X(\varnothing) = \mathbb{P}(X^{-1}(\varnothing)) = \mathbb{P}(\varnothing) = 0$
- $\mathbb{P}_X(B_1 \cup B_2)$ for $B_1 \cap B_2 = \varnothing$ is shown by noticing $X^{-1}(B_1 \cup B_2) = X^{-1}(B_1) \cup X^{-1}(B_2)$

6.3 Probability Distribution: Example of Coin Tossing

Consider a fair coin toss. The sample space is $\Omega = \{H, T\}$. Let X be a random variable representing the number of heads. The probability distribution of X is:

$$\mathbb{P}(X = 1) = \frac{1}{2}, \quad \mathbb{P}(X = 0) = \frac{1}{2}.$$

More generally, with n coin tosses with bias $p \in [0, 1]$

- $\Omega = \{0, 1\}^n$
- $\mathcal{F} = 2^\Omega$
- $\mathbb{P}$: Bernoulli: $\mathbb{P}(\{w\}) = p^{\#1}(1-p)^{\#0}$
- $X : \Omega \to \mathbb{R}$; $X(\omega) = \#1$'s in ω
- $\implies X \sim \text{Binomial}(n, p)$

7 Cumulative Distribution Function (CDF)

The CDF of a random variable X is defined as:

$$F_X(x) = \mathbb{P}(X \leq x).$$

It is a non-decreasing function that satisfies $\lim_{x \to -\infty} F_X(x) = 0$ and $\lim_{x \to \infty} F_X(x) = 1$. Also note, $F_X(x)$ is monotonically increasing and $F_X(x)$ is right-continuous.

8 Independent Random Variables

Random variables $X : \Omega \to \mathbb{R}$ and $Y : \Omega \to \mathbb{R}$ are independent if and only if $X^{-1}((-\infty, x])$ and $Y^{-1}((-\infty, y])$ are independent for all $x, y \in \mathbb{R}$.

In terms of the joint distribution, X and Y are independent if:

$$\mathbb{P}(X \leq x, Y \leq y) = F_X(x) \cdot F_Y(y)$$

where $\mathbb{P}(X \leq x, Y \leq y) = F_{(X,Y)}(x, y)$ is the joint cumulative distribution function (CDF) of (X, Y), and $F_{(X,Y)}$ is the CDF of the random vector $(X, Y) : \Omega \to \mathbb{R}^2$.

9 Expectation and Variance

9.1 Expectation

Consider a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and a random variable $X : \Omega \to \mathbb{R}$. The expectation or mean of X is defined as:

$$\mathbb{E}[X] = \int_{\Omega} X(\omega)\, d\mathbb{P}(\omega) \qquad \text{(abstract or Lebesgue integral)}$$

For discrete random variables,

$$\mathbb{E}[X] = \sum_{\omega \in \Omega} X(\omega)\, P_X(\omega)$$

where $P_X(\omega)$ is the probability mass function (pmf) of X.

For continuous random variables,

$$\mathbb{E}[X] = \int_{\Omega} X(\omega)\, f_X(\omega)\, d\omega$$

where $f_X(\omega)$ is the probability density function (pdf) of X.

More generally, for a function $g(X)$,

$$\mathbb{E}[g(X)] = \int_{\Omega} g(X(\omega))\, d\mathbb{P}(\omega)$$

For example, $g(X) = X^2$ and $\mathbb{E}[X^2]$ is the second moment of X.

Some properties of Expectation:

- **Linearity:** The expectation operator is linear,

$$\mathbb{E}\left[\sum_{i=1}^{d} \alpha_i X_i\right] = \sum_{i=1}^{d} \alpha_i \mathbb{E}[X_i]$$

- **Independence:** If X and Y are independent, then

$$\mathbb{E}[XY] = \mathbb{E}[X] \cdot \mathbb{E}[Y]$$

- **Equal Distributions:** If $\mathbb{P}_X = \mathbb{P}_Y$, then

$$\mathbb{E}[X] = \mathbb{E}[Y]$$

- **Ordering:** If $X \leq Y$ almost surely, i.e.,

$$\mathbb{P}(\{\omega \in \Omega \mid X(\omega) \leq Y(\omega)\}) = 1$$

  then

$$\mathbb{E}[X] \leq \mathbb{E}[Y]$$

9.2 Variance

The variance of X is defined as:

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

Hence,

$$\mathrm{Var}(X) \leq \mathbb{E}[X^2]$$

Standard Deviation: The standard deviation of X is

$$\mathrm{Std}(X) = \sqrt{\mathrm{Var}(X)}$$

Some properties of Variance:

- **Scaling:** For a constant α,
$$\mathrm{Var}(\alpha X) = \alpha^2 \mathrm{Var}(X)$$
- **Independence:** If X and Y are independent, then
$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$$

10 Moment Generating Function (MGF)

The MGF of a random variable X is defined as:

$$M_X(t) = \mathbb{E}[e^{tX}], \quad t \in \mathbb{R}.$$

11 Integral Identity

We begin with a useful integral identity that can help in understanding expected values. The expected value of X can be equivalently expressed using integration by parts as follows:

$$\mathbb{E}[X] = \int_0^\infty \mathbb{P}(X > a)\, da - \int_{-\infty}^0 \mathbb{P}(X < a)\, da.$$

Thus, for a non-negative random variable X, the expectation can also be expressed as:

$$\mathbb{E}[X] = \int_0^\infty \mathbb{P}(X > x)\, dx.$$

For a discrete random variable X, this integral can be replaced by a sum:

$$\mathbb{E}[X] = \sum_{i=1}^\infty \mathbb{P}(X = x_i), \tag{1}$$

12 Useful Inequalities

12.1 Union Bound

For any countable sequence of events $A_1, A_2, \ldots$, the union bound states:

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} \mathbb{P}(A_i).$$

Proof: The proof follows from the additivity axiom and the fact that the probability measure is non-negative.

12.2 Jensen's Inequality

For a convex function f and a random variable X:

$$\phi(\mathbb{E}[X]) \leq \mathbb{E}[f(X)].$$

For a concave function f, the inequality reverses:

$$\mathbb{E}[f(X)] \leq f(\mathbb{E}[X]).$$

Proof: Jensen's inequality is a consequence of the convexity of ϕ, which implies that the line segment joining any two points on the graph of ϕ lies above the graph.

12.3 Markov's Inequality

For a non-negative random variable X and $a > 0$:

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

Proof: Note that $X = X\mathbb{1}_{X \geq a} + X\mathbb{1}_{X<a}$. Thus,

$$\mathbb{E}[X] = \mathbb{E}[X\mathbb{1}_{X \geq a}] + \mathbb{E}[X\mathbb{1}_{X<a}].$$

Since $X\mathbb{1}_{X<a} \leq a$, we have

$$\mathbb{E}[X\mathbb{1}_{X<a}] \leq a\mathbb{P}(X < a).$$

Therefore,

$$\mathbb{E}[X] = \mathbb{E}[X\mathbb{1}_{X \geq a}] + \mathbb{E}[X\mathbb{1}_{X<a}] \geq \mathbb{E}[X\mathbb{1}_{X \geq a}] \geq a\mathbb{P}(X \geq a),$$

which gives

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

12.4 Chebyshev's Inequality

For any random variable X with finite variance and any $k > 0$:

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq k) \leq \frac{\mathrm{Var}(X)}{k^2}.$$

Proof: Apply Markov's inequality to $Y = (X - \mathbb{E}[X])^2$.

13 Law of Large Numbers and Central Limit Theorem

13.1 Strong Law of Large Numbers (SLLN)

If $X_1, X_2, \ldots$ are i.i.d. random variables with $\mathbb{E}[X_i] = \mu$, then:

$$S_n/n := \frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{a.s.} \mu.$$

If $\mathrm{Var}(X_i) = \sigma^2$ exists, then

$$\frac{\mathrm{Var}(S_N)}{N} \to \sigma^2.$$

13.2 Central Limit Theorem (CLT)

If $X_1, X_2, \ldots$ are i.i.d. random variables with $\mathbb{E}[X_i] = \mu$ and $\mathrm{Var}(X_i) = \sigma^2 < \infty$, then:

$$\frac{\sum_{i=1}^{n} X_i - n\mu}{\sqrt{n}\sigma} \xrightarrow{d} \mathcal{N}(0,1).$$

where $\xrightarrow{d}$ denotes convergence in distribution to the standard normal distribution.

14 Conditional Expectation

Conditional expectation is a fundamental concept in probability theory, particularly useful when dealing with random variables and their dependencies. The general definition of conditional expectation can be complex and often requires measure theory for a rigorous understanding. However, we will start with a simpler, more intuitive approach before gradually building up to more general cases.

14.1 Definition and Basic Properties

Consider a random variable X. The regular expectation of X can be expressed as:

$$\mathbb{E}[X] = \int x \, d\mathbb{P}(x) \tag{2}$$

where the expectation is taken with respect to the probability measure $\mathbb{P}$. For discrete random variables, this can be written as:

$$\mathbb{E}[X] = \sum_i x_i \, \mathbb{P}(X = x_i) \tag{3}$$

Now, instead of the regular expectation, we can consider the expectation conditioned on an event A. The conditional expectation of X given A is defined as:

$$\mathbb{E}[X \mid A] = \sum_i x_i \, \mathbb{P}(X = x_i \mid A) \tag{4}$$

Similarly, the conditional expectation of X given a discrete random variable Y taking value y is:

$$\mathbb{E}[X \mid Y = y] = \sum_i x_i \, \mathbb{P}(X = x_i \mid Y = y) \tag{5}$$

For continuous random variables, the conditional expectation involves the conditional probability density function (PDF), which is defined as the ratio of the joint PDF to the marginal PDF of Y:

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x, y)}{f_Y(y)} \tag{6}$$

The conditional expectation of X given Y in this case is:

$$\mathbb{E}[X \mid Y = y] = \int x \, f_{X|Y}(x \mid y) \, dx \tag{7}$$

14.2 General Definition

In general, the conditional expectation of a random variable X with respect to another random variable Y is defined as any function $g(Y)$ that satisfies:

$$\mathbb{E}[X \mid \mathcal{G}] = g(Y) \tag{8}$$

where $\mathcal{G}$ is the σ-algebra generated by Y ($\mathcal{G} = \sigma(Y) := \{Y^{-1}(A) \mid A \in B(\mathbb{R})\}$ is the σ-algebra generated by Y.). This σ-algebra essentially contains all the information needed to know Y deterministically.

15 Important Properties of Conditional Expectation

15.1 Independence

If X and Y are independent random variables, then the conditional expectation of X given Y is simply the expectation of X:

$$\mathbb{E}[X \mid Y] = \mathbb{E}[X] \tag{9}$$

This result is intuitive since knowing Y does not provide any additional information about X when X and Y are independent.

15.2 σ-Algebra Measurability

If X is $\sigma(Y)$-measurable, meaning that X can be treated as a function of Y, then the conditional expectation of X given Y is X itself:

$$\mathbb{E}[X \mid Y] = X \tag{10}$$

This is because, given Y, X is deterministic and does not vary randomly.

15.3 Tower Property of Conditional Expectation

The tower property, or the law of total expectation, allows us to express the expectation of a random variable in terms of its conditional expectations. Specifically, for random variables X and Y:

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]] \tag{11}$$

This property can be extended to incorporate multiple random variables:

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[\mathbb{E}[X \mid Z] \mid Y]] \tag{12}$$

15.4 Examples

15.4.1 Example 1: Random Sum of Random Variables

Consider a random variable Y, defined as the sum of N random variables X_i, where X_i are i.i.d. Bernoulli random variables with parameter p, and N itself is a random variable independent of the X_i's. The goal is to find $\mathbb{E}[Y]$.

First, we express Y as:

$$Y = \sum_{i=1}^{N} X_i \tag{13}$$

If N were deterministic, we could write:

$$\mathbb{E}[Y] = \sum_{i=1}^{N} \mathbb{E}[X_i] = Np \tag{14}$$

However, since N is random, the correct approach is to use the law of total expectation:

$$\mathbb{E}[Y] = \mathbb{E}\left[\mathbb{E}\left[\sum_{i=1}^{N} X_i \mid N\right]\right] \tag{15}$$

By conditioning on N, the inner expectation becomes:

$$\mathbb{E}\left[\sum_{i=1}^{N} X_i \mid N\right] = Np \tag{16}$$

Thus:

$$\mathbb{E}[Y] = p\,\mathbb{E}[N] \tag{17}$$

Alternatively, we can solve this problem using the indicator function:

$$Y = \sum_{i=1}^{\infty} X_i \mathbb{I}\{N \geq i\} \tag{18}$$

Taking the expectation and leveraging the independence of X_i and N, we obtain:

$$\mathbb{E}[Y] = p \sum_{i=1}^{\infty} \mathbb{P}(N \geq i) = p\,\mathbb{E}[N] \tag{19}$$

15.4.2 Example: Conditional Expectation and Martingales

Consider now a sequence of random variables Y_n defined as:

$$Y_n = \prod_{i=1}^{n} \frac{f(X_i)}{g(X_i)}, \tag{20}$$

where f and g are probability density functions (PDFs), and $X_i \sim g$ in an iid fashion. The goal is to find $\mathbb{E}[Y_n \mid Y_1, \ldots, Y_{n-1}]$.

First, express the conditional expectation as:

$$\mathbb{E}[Y_n \mid Y_1, \ldots, Y_{n-1}] = \prod_{i=1}^{n-1} \frac{f(X_i)}{g(X_i)} \cdot \mathbb{E}\left[\frac{f(X_n)}{g(X_n)} \mid Y_1, \ldots, Y_{n-1}\right]. \tag{21}$$

Given that X_n is independent of $X_1, \ldots, X_{n-1}$, the conditional expectation simplifies to:

$$\mathbb{E}\left[\frac{f(X_n)}{g(X_n)} \mid Y_1, \ldots, Y_{n-1}\right] = \int \frac{f(x)}{g(x)} g(x)\, dx = \int f(x)\, dx = 1, \tag{22}$$

so:

$$\mathbb{E}[Y_n \mid Y_1, \ldots, Y_{n-1}] = Y_{n-1}. \tag{23}$$

This result implies that $\{Y_n\}$ is a martingale, a concept that will be useful in our discussions of high-probability guarantees and theoretical analysis of optimization algorithms.

16 Filtration

A filtration $\mathbb{F} = \{\mathcal{F}_t\}_{t \geq 0}$ is an increasing sequence of sigma algebras, i.e., $\mathcal{F}_s \subseteq \mathcal{F}_t$ for $s \leq t$, representing the evolution of information over time. In a probabilistic setting, $\mathcal{F}_t$ typically represents the information available up to time t. For example, in a stochastic process $\{X_t\}_{t \geq 0}$, the natural filtration is given by:

$$\mathcal{F}_t = \sigma(X_s : 0 \leq s \leq t),$$

where $\sigma(X_s : 0 \leq s \leq t)$ is the smallest sigma algebra containing all events determined by the process $\{X_s\}_{0 \leq s \leq t}$.

Example: Define $\mathcal{F}_n := \sigma(X_k \mid k \leq n)$, the σ-algebra generated by the random variables X_k for $k \leq n$.

- In our context, let $\mathcal{F}_n := \sigma(z_1, \ldots, z_n)$, which is the σ-algebra generated by the samples (mini-batches) in iterations $1, \ldots, n$.
- A sequence $\{X_i\}$ is said to be $\mathbb{F}$-adapted if X_t is $\mathcal{F}_t$-measurable.
- A sequence $\{X_i\}$ is said to be $\mathbb{F}$-predictable if X_t is $\mathcal{F}_{t-1}$-measurable.

17 Martingale Difference

Given a filtration $\{\mathbb{F} = \mathcal{F}_t\}_{t \geq 0}$ and a stochastic process $\{X_t\}_{t \geq 0}$ adapted to this filtration (i.e., X_t is $\mathcal{F}_t$-measurable), the process is a *martingale* if:

1. $\mathbb{E}[|X_t|] < \infty$ for all $t \geq 0$,
2. $\mathbb{E}[X_{t+1} \mid \mathcal{F}_t] = X_t$ almost surely.

The difference sequence $D_t = X_{t+1} - X_t$ is called the *martingale difference sequence*. The key property is that:

$$\mathbb{E}[D_t \mid \mathcal{F}_t] = 0 \text{ almost surely.}$$

18 Azuma-Hoeffding Inequality

The Azuma-Hoeffding inequality provides a concentration bound for the sum of bounded martingale differences. Let $\{X_t\}_{t=0}^n$ be a martingale with respect to a filtration $\{\mathcal{F}_t\}_{t=0}^n$, and assume that $|X_t| \leq b_t$. Then, with probability at least $1-\delta$, it holds that

$$\sum_{t=1}^{T} X_t \leq \sqrt{2 \log\left(\frac{1}{\delta}\right) \sum_{i=1}^{T} b_i^2}.$$

This inequality is particularly useful for deriving high probability bounds on the deviation of a martingale from its expected value.